

Original Article

Anthocyanins and flavonoids derived from *Clitoria ternatea* L. flower inhibit bladder cancer growth via suppressing fatty acid synthesis mediated by SREBP1 pathway

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Abstract

Clitoria ternatea L. flowers are used as traditional herbal medicines and are known for their advanced pharmacological activities. Flavonoids and anthocyanins reportedly contribute to the therapeutic properties of *C. ternatea* flowers; however, their potential anti-bladder cancer effects and molecular mechanisms remain unknown. In this study, flavonoid- and anthocyanin-rich samples from *C. ternatea* flowers (DDH) are prepared via macroporous resin-based extraction coupled with an efficient and reliable two-dimensional UPLC-DAD-MS/MS method. *In vitro* and *in vivo* studies reveal that DDH can inhibit bladder cancer cell growth and enhance the anti-bladder cancer activity of cisplatin. RNA-seq combined with KEGG analysis reveals that fatty acid synthesis is closely related to the anti-bladder cancer effect of DDH. Furthermore, DDH dose-dependently reduces cellular fatty acid levels in bladder cancer cells, and the addition of fatty acids significantly mitigates DDH-induced cell growth inhibition. Subsequent findings reveal that DDH downregulates sterol regulatory element-binding protein 1 (SREBP1), a key transcriptional regulator of *de novo* fatty acid synthesis in cancer cells, and its downstream targets (FASN, SCD1, and ACC). Additionally, this study demonstrates that gallic acid not only enhances the stability of DDH but also synergistically potentiates its anti-bladder cancer activity. Our study suggests that targeting the SREBP1 pathway is an effective strategy in bladder cancer therapy, and the ability of DDH to induce cell death by inhibiting the SREBP1 pathway and its good tolerance in mice make it a promising strategy for preventing and treating bladder cancer.

Key words combination therapy, bladder cancer, extraction optimization, lipogenesis, phytochemicals, two-dimensional UPLC-DAD-MS/MS

Introduction

The incidence and mortality rates of bladder cancer, a malignant urological tumor, are increasing annually. The Global Cancer Statistics 2022 report estimated that there were 614,000 new cases and 220,000 deaths worldwide associated with bladder cancer in 2022 [1]. A prominent metabolic alteration observed in cancer is the reprogramming of lipid metabolism. Lipids not only are important components of biological membranes but are also involved in cellular energy metabolism and storage as well as in intracellular signaling [2]. Tumor cells increase lipid metabolism,

particularly fatty acid synthesis, to meet their needs for rapid proliferation, survival, migration, invasion, and metastasis [3]. Sterol regulatory element-binding protein 1 (SREBP1) is a key transcriptional regulator of *de novo* fatty acid synthesis in cells and regulates the expressions of key enzymes involved in lipid synthesis, such as stearyl-CoA desaturase 1 (SCD1), fatty acid synthase (FASN), and acetyl-CoA carboxylase (ACC) [4]. Emerging evidence indicates that inhibiting the SREBP1 pathway can suppress intracellular fatty acid synthesis, thereby reducing the proliferation of tumor cells [3,4]. The abnormal hyperactivity of

lipid metabolism in bladder cancer is closely linked to its development, progression, and drug resistance [5–7]. Hence, targeting SREBP1 pathway-mediated fatty acid synthesis holds promise as a strategy for treating bladder cancer.

Clitoria ternatea L. is a common plant in the Sipsongpanna region of Yunnan, China, and its blue flowers are known for their advanced health benefits, such as anti-inflammatory, antioxidant, and anti-obesity effects [8–10]. The primary active constituents of *C. ternatea* flowers are flavonoids and anthocyanins [11–13]. Reportedly, flavonoids and anthocyanins from plant sources are promising agents for inhibiting the proliferation of various tumor cells, such as hepatocellular carcinoma, lung cancer, and bladder cancer [14,15]. However, whether flavonoids and anthocyanins from *C. ternatea* flowers can inhibit the proliferation of bladder cancer cells as well as their underlying mechanism of action remain unclear.

In this study, we developed a fast and efficient two-dimensional UPLC–DAD–MS/MS analytical method for simultaneous qualitative and quantitative analyses of flavonoids and anthocyanins in *C. ternatea* flowers. We optimized a macroporous resin-based approach to extract flavonoid- and anthocyanin-rich samples from *C. ternatea* flowers (DDH). Our results indicate that DDH could inhibit the growth of bladder cancer cells and enhance the anticancer effects of cisplatin (CDDP). Further mechanistic studies revealed that DDH kills bladder cancer cells by inhibiting the SREBP1 pathway and reducing the level of cellular fatty acids. Additionally, we found that gallic acid (GAD) not only improves the stability of DDH but also synergistically potentiates its anticancer activity against bladder cancer *in vitro* and *in vivo*. Therefore, DDH shows promise as a therapeutic agent for bladder cancer.

Materials and Methods

Chemicals and reagents

Cyanidin-3-glucoside (purity $\geq 98\%$), quercetin-3-rutinoside (purity $\geq 98\%$), and GAD (purity $\geq 98\%$) were purchased from Chengdu DESITE BioTechnology Co., Ltd. (Chengdu, China). *C. ternatea* flowers were purchased from a local market in Kunming, China, and herbarium samples (No. 2023-3-7-Diedouhua) were deposited at the Key Laboratory of Target Drug Screening and Utilization for Universities in Yunnan Province. Formic acid (HPLC grade) was purchased from Tianjin Kermel Chemical Reagent Co., Ltd. (Tianjin, China). Acetonitrile and trifluoroacetic acid (HPLC grade) were purchased from Fisher Chemicals Co., Ltd. (Fair Lawn, USA). Macroporous resins LX-32, LX-68, XDA-6 and XDA-7 were purchased from Xi'an Sunresin New Material Co., Ltd. (Xi'an, China); AB-8 was provided by Cangzhou Bon Adsorber Technology Co., Ltd. (Cangzhou, China); X-5 was purchased from the Chemical Plant of Nankai University (Tianjin, China); and LS-302 was purchased from Shaanxi Lanshen Special Resin Co., Ltd. (Xi'an, China). Oleic acid (OA) and palmitoleic acid (POA) were purchased from MedChemExpress (Monmouth Junction, USA). Antibodies against SREBP1 (ab28481), FASN (ab128870) and SCD1 (ab19862) were purchased from Abcam (Cambridge, UK); antibodies against ACC (3662) were purchased from Cell Signaling Technology (Danvers, USA); and antibodies against GAPDH (200306-7E4), goat anti-rabbit IgG (511203) and rabbit anti-mouse IgG (701051) were purchased from ZEN-BIOSCIENCE (Chengdu, China). All other chemicals and reagents were of analytical grade, and deionized

water was used in all the experiments.

Extraction and separation of anthocyanins and flavonoids from *C. ternatea* flowers

The extraction of total anthocyanins (TA) and total flavonoids (TF) from *C. ternatea* flowers was optimized via response surface methodology and central composite design (Supplementary Table S1), and macroporous resin was used to separate TA and TF from the *C. ternatea* flower crude extract. A detailed description is provided in the Supplementary Material.

First-dimensional UPLC–DAD analysis

UPLC analysis of *C. ternatea* flower anthocyanins and flavonoids was performed on an Agilent HPLC series 1290 system (Agilent Technologies Co., Ltd., Santa Clara, USA) consisting of a quaternary gradient pump, a thermostatic column compartment, an autosampler, a diode array detector and a ZORBAX Eclipse Plus C18 (4.6 mm $\times$ 150 mm, 3.5 μ m) analytical column. The mobile phase consisted of acetonitrile (A) and an acidic aqueous solution (0.1% trifluoroacetic acid, v/v) (B). The gradient elution program was as follows: 0–10 min, 88%–82% B; 10–20 min, 82%–78% B; and 20–22 min, 78%–0% B. The injection volume was 5 μ L, the flow rate was 1.5 mL/min, the detection wavelengths were 360 nm and 525 nm, and the column temperature was 35°C.

Second-dimensional UPLC–MS analysis

UPLC–MS analysis of *C. ternatea* flower anthocyanins and flavonoids was performed with an Agilent 1290 UPLC system equipped with an Agilent ZORBAX Eclipse Plus C18 column (100 mm $\times$ 4.5 mm, 3.5 mm particle size) and an Agilent 6460C triple quadrupole mass spectrometer with an AJS electrospray ionization source (Agilent Technologies). The mobile phases were acetonitrile and formic acid aqueous solution (1%, v/v) at a 10:90 (v/v) ratio, the flow rate was 0.5 mL/min, and the injection volume was 20 μ L. The MS conditions used were as follows: positive mode for anthocyanins and negative mode for flavonoids; sheath gas temperature, 350°C; sheath gas flow rate, 12.0 L/min; capillary voltage, 3500 V; nebulizer pressure, 45 psi; scan range, 100–2000 m/z; drying gas temperature, 300°C; and flow rate, 6.0 L/min.

Cell culture

The human bladder cancer cell lines T24, UM-UC-3 and J82 were purchased from CellCook (Guangzhou, China). T24 cells were cultured in RPIM 1640 medium containing 10% (v/v) fetal bovine serum (CellCook), UM-UC-3 and J82 cells were cultured in MEM containing 10% (v/v) fetal bovine serum (CellCook), and all three cell lines were incubated at 37°C in a 5% CO₂ atmosphere.

Anti-proliferative assay

Bladder cancer cells were seeded in 96-well plates (5 $\times$ 10³ cells/well) and cultured for 24 h, and the cells were treated with DDH for 48 h. An MTT assay was used to determine cell viability according to previously described methods [16]. Briefly, bladder cancer cells were incubated with MTT solution (20 μ L; 5 mg/mL; Solarbio, Beijing, China) for 2 h at 37°C, and 100 μ L of DMSO was added to each well. After mixing thoroughly, the absorbance was measured at 570 nm. In addition, the morphology of DDH-treated (12 h) bladder cancer cells was analyzed, and a colony formation assay was performed according to previously described methods [16]. Briefly,

Table 1. Identification of anthocyanins and flavonoids via a two-dimensional UPLC-DAD-MS-MS method

First-dimensional UPLC-DAD				Second-dimensional UPLC-MS		
Peaks	Retention time (min)	UV/Vis	Percentage area ^a	MS ^b	MS/MS ^b	Tentative identification
Anthocyanins						
A1	3.417	196, 284, 536	1.36 %	1405	303	Delphinidin derivative
A2	3.963	196, 288, 542	2.35 %	1491	303	Ternatin A3
A3	7.147	222, 290, 542	3.14 %	1329	303	Ternatin C1
A4	7.810	224, 288, 552	7.18 %	1799	303	Ternatin A2
A5	9.387	198, 266, 548	3.36 %	1329	303	Ternatin B4
A6	9.800	226, 290, 548	4.04 %	1638	303	Ternatin B3
A7	13.233	222, 290, 550	14.93 %	1638	303	Ternatin B2
A8	13.740	224, 290, 552	17.86 %	1945	303	Ternatin B1
A9	15.400	226, 294, 550	10.51 %	1475	303	Ternatin D2
A10	18.267	226, 292, 553	16.99 %	1783	303	Ternatin D1
A11	18.833	230, 292, 548	2.55 %	1183	303	Delphinidin derivative
Flavonoids						
F1	5.940	202, 256, 354	4.92 %	755	301	Quercetin-3-2 ^G -rhamnosylrutinoside
F2	7.223	204, 256, 354	9.96 %	609	301	Quercetin-3-neohesperidoside
F3	7.660	198, 264, 346	16.96 %	739	285	Kaempferol-3-2 ^G -rhamnosylrutinoside
F4	8.723	204, 256, 354	4.41 %	609	301	Quercetin-3-rutinoside
F5	9.207	198, 264, 348	47.13 %	593	285	Kaempferol-3-neohesperidoside
F6	9.540	204, 256, 354	4.03 %	741	301	Quercetin derivative
F7	12.237	226, 266, 346	4.59 %	635	285	Kaempferol derivative

^aThe detection wavelengths for anthocyanins and flavonoids were 525 nm and 360 nm, respectively. ^bThe MS scan conditions for anthocyanins and flavonoids were positive and negative, respectively.

bladder cancer cells were seeded evenly in 6-well plates (1000 cells/well) and cultured for 24 h and then treated with medium containing DDH for 12 days. The colonies were fixed with 4% paraformaldehyde and stained with Giemsa stain for later analysis.

Human bladder cancer xenograft models

Wild-type AB strain zebrafish embryos were purchased from Hunter Biotech (Hangzhou, China). T24 cells labeled with CM-DiI red fluorescence (200 cells/tail) were transplanted into the yolk sac of wild-type AB strain zebrafish (2 days after insemination) to establish a bladder cancer xenograft zebrafish model, and 30 tails/well were randomly assigned to 6-well plates. These zebrafish were treated with DDH (15 µg/mL and 60 µg/mL), CDDP (15 µg/mL), or DDH (15 µg/mL) + CDDP (15 µg/mL) at 35°C for 2 days, and the red fluorescence intensity was calculated to determine the anti-bladder cancer effect of DDH in the zebrafish xenograft model.

BALB/c nu/nu male mice were purchased from the Animal Research and Resource Center at Yunnan University (Kunming, China), and a bladder cancer xenograft nude mouse model was established as previously described [17]. Nude mice bearing T24 xenografts were gavaged with vehicle, DDH (200 mg/kg), GAD (50 mg/kg), or DDH (200 mg/kg) + GAD (50 mg/kg) once a day. At the end of the study, the tumors, livers and kidneys were harvested from the sacrificed mice (carbon dioxide narcosis) for subsequent assays.

All animal experiments were performed according to the Animal

Ethics Procedures and approved by the Yunnan University Laboratory Animal Welfare Ethics Committee (ethics approval/permit number: YNU20230667).

Western blot analysis

Bladder cancer cells (T24, UM-UC-3 and J82) were seeded in 6-well plates (2 × 10⁵ cells/well) and cultured for 24 h, and the cells were treated with DDH for 12 h. Western blot analysis was performed according to previously published protocols [16]. Briefly, the cells were lysed in RIPA buffer, and the lysates were centrifuged at 12,000 g for 30 min at 4°C. The cell lysates were boiled at 95°C for 10 min after being mixed with sample buffer. The mixture was resolved by SDS-PAGE and transferred to a PVDF membrane. The membrane was incubated with primary and secondary antibodies and detected via an enhanced chemiluminescence detection kit (Thermo Fisher Scientific, Waltham, USA).

RNA-seq analysis

T24 cells were seeded in culture dishes (10 cm in diameter; 8 × 10⁶ cells/dish) and cultured for 24 h, and then treated with DDH (200 mg/mL) for 12 h. The cells were washed with PBS (4°C) 3 times and collected via centrifugation (100 g). The RNA-seq analysis was performed by PANOMIX Biomedical Tech Co., Ltd. (Suzhou, China). The DEGs between the DDH group and the control group were compared via DESeq software (|log2fold change| > 1, P < 0.05), and the Kyoto Encyclopedia of Genes and Genomes (KEGG) database and KAAS software (Bi-directional Best Hit) were used for KEGG

pathway analysis.

Fatty acid analysis

Cellular free fatty acids were measured using an Amplex Red Free Fatty Acid Test Kit (Beyotime Biotechnology, Shanghai, China) following the manufacturer's protocol. Briefly, bladder cancer cells in 6-well plates were treated with DDH for 12 h and recovered via isopropanol (4°C). The samples were mixed with working solution provided by the manufacturer and incubated for 30 min at 37°C (in the dark). After mixing thoroughly, the absorbance was measured at 570 nm.

Thermal stability assay

The thermal stabilities of TA and TF were determined according to previously described methods [18]. Briefly, DDH (0.4 mg/mL) and GAD (0.2, 0.4 and 0.8 mg/mL) were dissolved in 100 mL of the Mizone beverage. The model beverages were heated in a water bath at 90°C and analyzed every 30 min. The degradation of TA and TF in the model beverages was evaluated via a first-order model [18].

Statistical analysis

Response surface experiments were designed and analyzed via Design Expert 13 software. Statistical analyses were performed via Prism 8.3.0 (GraphPad Software, La Jolla, USA). One-way analysis of variance and Tukey's multiple comparison test were used to determine the statistical significance. $P < 0.05$ was considered statistically significant.

Results

Quantitative and qualitative analysis of anthocyanins and flavonoids from *C. ternatea* flowers

A UPLC–DAD method was developed to analyze anthocyanins and flavonoids from *C. ternatea* flower (Figure 1A,B). The linear calibration curves for TA and TF were determined via cyanidin-3-glucoside and quercetin-3-rutinoside standard solutions, respectively. Method validation was performed according to a previous study [19], and a detailed description is provided in the Supplementary Material. The results indicate that the method is both valid and efficient for analyzing anthocyanins and flavonoids from *C. ternatea* flowers.

We identified 11 anthocyanins and 7 flavonoids (percentage area $> 1\%$ in the first-dimensional UPLC-DAD chromatogram) from *C. ternatea* flowers via a second-dimensional method (Figure 1C and Table 1). The anthocyanins found in *C. ternatea* flowers are delphinidin derivatives ($MS/MS = 303\ m/z$) (Table 1). Four main peaks (A7, 14.93%; A8, 17.86%; A9, 10.51%; A10, 16.99%) were identified as Ternatin B2 ($MS^+ = 1638\ m/z$; $MS/MS = 303\ m/z$), Ternatin B1 ($MS^+ = 1945\ m/z$; $MS/MS = 303\ m/z$), Ternatin D2 ($MS^+ = 1475\ m/z$; $MS/MS = 303\ m/z$), and Ternatin D1 ($MS^+ = 1783\ m/z$; $MS/MS = 303\ m/z$) [11,12,20]. Peak A4 ($MS^+ = 1799\ m/z$; $MS/MS = 303\ m/z$) was identified as Ternatin A2 [21]. Peak A6 ($MS^+ = 1638\ m/z$; $MS/MS = 303\ m/z$) exhibited a fragmentation pattern similar to that of anthocyanin A7, with a lower content and retention time than those of anthocyanin A7. Therefore, on the basis of previous reports, anthocyanin A6 was identified as Ternatin B3 [21,22]. As natural anthocyanin compounds often have structural diversity, isomers are common. Ternatin C1 and Ternatin B4 had similar fragmentation patterns ($MS^+ = 1799\ m/z$; $MS/MS = 303\ m/z$); however, the retention time and maximum absorption of Ternatin

C1 were lower than those of Ternatin B4 [13]. Thus, peaks A3 and A5 were tentatively identified as Ternatin C1 and Ternatin B4, respectively. Additionally, peak A2 was identified as Ternatin A3 [13]. Anthocyanins A1 and A11, delphinidin derivatives ($MS/MS = 303\ m/z$), have not been previously reported, and their complete structures can be elucidated via nuclear magnetic resonance spectroscopy.

The primary flavonoid in *C. ternatea* flower is kaempferol-3-neohesperidoside [23]. Peak F5 exhibited a similar fragmentation pattern to that of kaempferol-3-neohesperidoside ($MS^+ = 593\ m/z$; $MS/MS = 285\ m/z$), as depicted in Table 1, and the main flavonoid peak was confirmed to be kaempferol-3-neohesperidoside [13]. The peak F3 content was second only to kaempferol-3-neohesperidoside (peak F5) and was identified as kaempferol-O-rhamnosylrutinoside ($MS^+ = 739\ m/z$; $MS/MS = 285\ m/z$), which is consistent with a previous report [22]. Peaks F2 and F4 exhibited similar mass spectrometry signals ($MS^+ = 609\ m/z$; $MS/MS = 301\ m/z$), with Peak F4 sharing a similar fragmentation pattern and retention time to the quercetin-3-rutinoside standard (Table 1 and Figure 1B). Thus, peaks F2 and F4 were identified as quercetin-3-rutinoside and quercetin-3-neohesperidoside, respectively [13,23]. Peaks F6 and F7 have not been previously reported and were provisionally identified as quercetin and kaempferol derivatives, respectively. Furthermore, peak F1 ($MS^+ = 755\ m/z$; $MS/MS = 301\ m/z$) was confirmed to be quercetin-3-2^G-rhamnosylrutinoside [12,13].

Extraction and separation of anthocyanins and flavonoids from *C. ternatea* flowers

The extraction parameters for TA and TF from *C. ternatea* flowers were optimized via response surface methodology (Supplementary Tables S1 and S2), and a detailed description is provided in the Supplementary Material. Both models were found to be significant ($P < 0.05$), and the lack of fit of each response was insignificant ($P > 0.05$) (Supplementary Table S2). Furthermore, the coefficients of determination (R^2) for both models were 86.8% and 89.2%, respectively (Figure 2A). These results suggest an excellent fit between the experimental and predicted values, indicating that the two mathematical models are statistically acceptable and adequate for predicting the extraction of TA and TF from *C. ternatea* flowers (Figure 2B). Given the complex effects of various extraction parameters on the extraction yield of TA and TF (Figure 2C,D), simultaneous optimization was performed via Design-Expert 13 software to maximize their yield from *C. ternatea* flowers. The optimum extraction conditions for TA and TF were identified as an extraction time of 33 min, an ethanol concentration of 45%, and an extraction temperature of 65°C, resulting in maximum yields of TA (3.89 mg/g) and TF (20.38 mg/g).

Seven different microporous resins (LX-32, LS-305, XDA-7, X-5, LX-68, XDA-6, and AB-8) were evaluated for their ability to separate TA and TF from the crude extract of *C. ternatea* flowers. Our results showed that LX-32 and LS-305 demonstrated higher TF adsorption rates than the other resins. However, the adsorption capacity of LS-305 for TA was significantly greater than that of LX-32 ($P < 0.05$; Figure 2E). Both LS-305 and LX-32 exhibited similar desorption rates for TA and TF ($P > 0.05$). There was no significant difference ($P > 0.05$) in the recovery rate of TF between the two resins (Figure 2E). In conclusion, LS-305 proved to be an effective microporous resin for the separation of TA and TF from the crude extract of *C. ternatea* flowers.

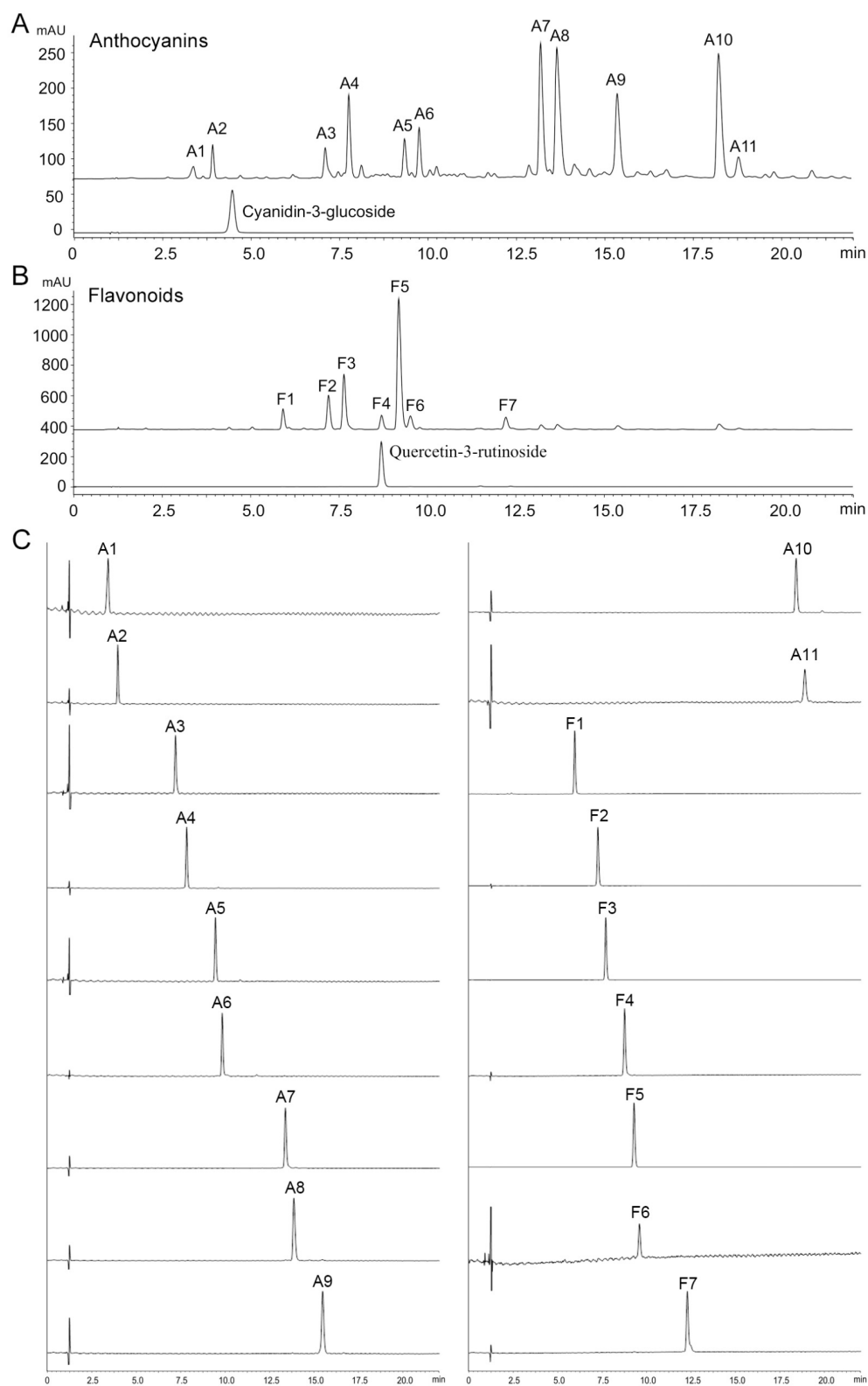


Figure 1. Two-dimensional UPLC-DAD-MS/MS analysis of anthocyanins and flavonoids from *C. ternatea* flowers (A) First-dimensional UPLC-DAD chromatograms of *C. ternatea* flower anthocyanins (A1–A11) and the cyanidin-3-glucoside standard. (B) First-dimensional UPLC-DAD chromatograms of *C. ternatea* flower flavonoids (F1–F7) and the quercetin-3-rutinoside standard. (C) Second-dimensional UPLC-MS analysis of anthocyanins (A1–A11) and flavonoids (F1–F7) from *C. ternatea* flowers.

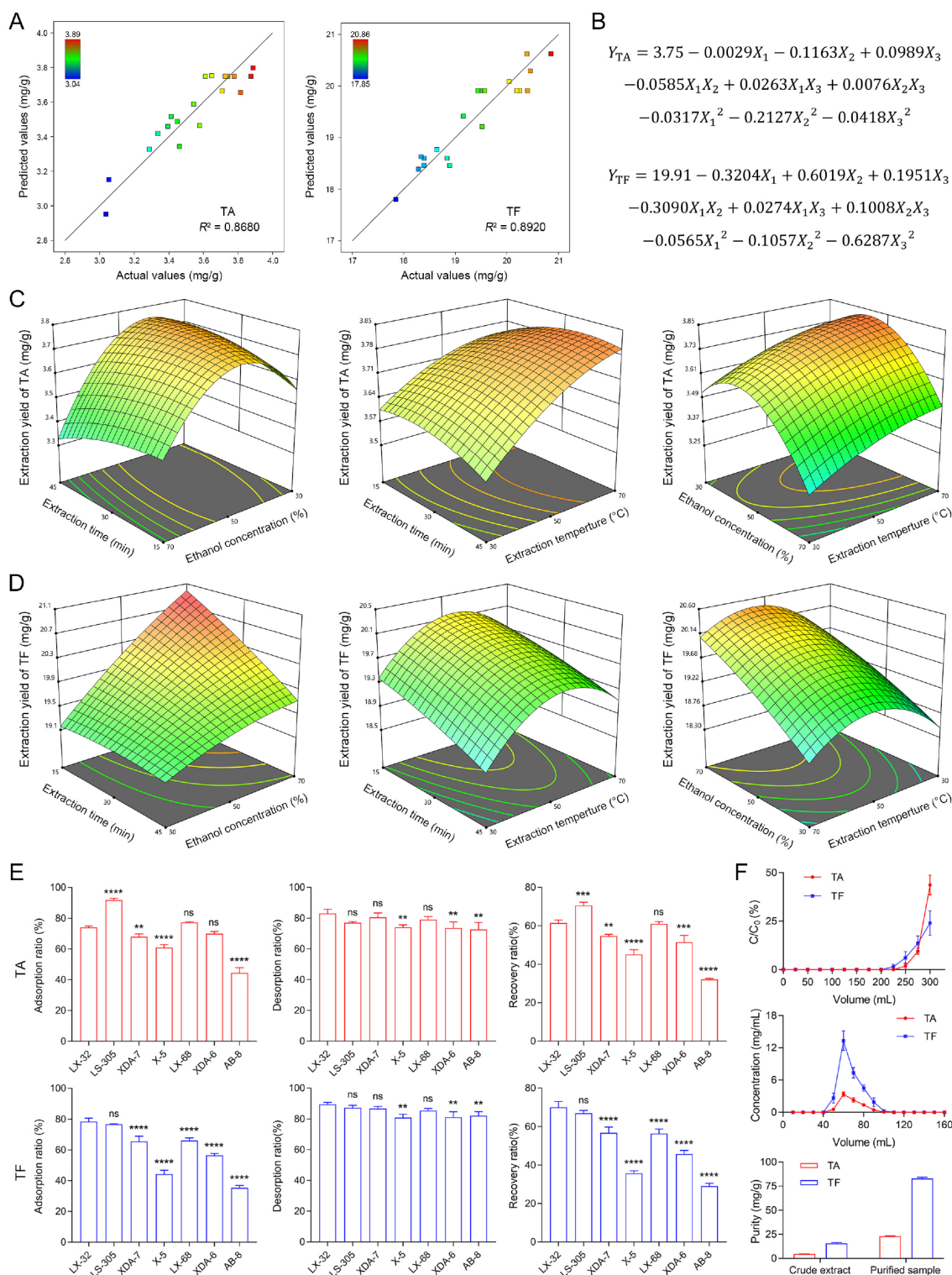


Figure 2. Preparation of DDH from *C. ternatea* flowers (A) Coefficients of determination (R^2) between predicted and actual values of TA or TF. (B) Mathematical models for predicting extraction conditions of TA and TF from *C. ternatea* flowers. (C) Response surface plots showing the combined effect of extraction temperature, extraction time and ethanol concentration on extraction yield of TA from *C. ternatea* flowers. (D) Response surface plots showing the combined effect of extraction temperature, extraction time and ethanol concentration on extraction yield of TF from *C. ternatea* flowers. (E) Adsorption ratio, desorption ratio and recovery ratio of studied macroporous resins. $n=3$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, and ns means not significant compared with LX-32 resin. (F) Dynamic breakthrough curves and desorption curves of TA and TF in a resin column of LS-305, and the purity of TA and TF in crude extract or purified sample (DDH).

The dynamic leakage curve was crucial for determining the maximum feed volume of the LS-305 resin (Figure 2F), and a detailed description is provided in the Supplementary Material. The crude extract of *C. ternatea* flowers contained 4.57 mg/g TA and 23.04 mg/g TF. Following purification with LS305 resin (Figure 2F), the contents of TA and TF increased to 15.635 and 82.82 mg/g, respectively, representing a greater than 3-fold increase. The purified anthocyanin- and flavonoid-rich DDH derived from *C. ternatea* flowers was used for antitumor experiments after lyophilization.

Antiproliferative effects of DDH on human bladder cancer cells

The anticancer properties of DDH were assessed in a panel of human bladder cancer cells. DDH reduced the viability of T24, UM-UC-3, and J82 cells in a dose-dependent manner, with IC_{50} values of 34.40, 41.44, and 52.72 μ g/mL, respectively (Figure 3A). Additionally, DDH affected the morphology of bladder cancer cells, leading to shrinkage, loose intercellular contact, and blurred cell contours (Figure 3B). Moreover, DDH reduced the colony formation of T24, UM-UC-3, and J82 cells, indicating the inhibition of long-term proliferation of bladder cancer cells (Figure 3C). Furthermore, DDH dose-dependently increased the cytotoxicity of CDDP, a first-line therapeutic for bladder cancer, to T24, UM-UC-3, and J82 cells (Figure 3D).

The therapeutic potential of DDH against bladder cancer was further investigated in a xenograft zebrafish model bearing T24 cells labeled with a red fluorescent dye. The therapeutic dosages of DDH were first screened for tolerability and toxicity in zebrafish after a 2-day treatment; the dosages that did not affect the survival rate of zebrafish were selected for further *in vivo* experiments (Figure 3E). DDH reduced the fluorescence density in a dose-dependent manner, indicating a promising anti-bladder cancer effect in the zebrafish model (Figure 3F,G). Compared with CDDP or DDH alone, CDDP (15 μ g/mL) inhibited tumor growth, and combined treatment with DDH (15 μ g/mL) and CDDP (15 μ g/mL) had a more potent antitumor effect (Figure 3F,G). In conclusion, DDH possessed antiproliferative properties and enhanced the antitumor effect of CDDP against bladder cancer.

DDH inhibits SREBP1-mediated lipogenesis in bladder cancer cells

To understand the underlying mechanism of action of DDH in bladder cancer cells, RNA-seq analysis was performed (Figure 4A). Our analysis revealed a total of 1896 differentially expressed genes (DEGs), with 980 genes being downregulated and 916 upregulated when DDH-treated cells were compared with control cells ($P < 0.05$; Figure 4A). Kyoto Encyclopedia of Genes and Genomes analysis indicated that DDH might have the ability to regulate lipid metabolism in bladder cancer cells, particularly by impacting processes such as the biosynthesis of unsaturated fatty acids, fatty acid biosynthesis, fatty acid degradation, sphingolipid metabolism, and glycosphingolipid biosynthesis-ganglio series (Figure 4A).

SREBP1, as a master transcriptional regulator of lipogenesis, promotes the transcription of target genes, including those encoding crucial enzymes involved in lipogenesis, such as FASN, SCD1, and ACC [24]. Elevated expression or activation of these lipogenic genes has been observed in various cancer types and is closely associated with the proliferation and survival of tumor cells [4]. Analysis of

SREBP1, FASN, ACC, and SCD1 in bladder cancer via the GEPIA database (<http://gepia.cancer-pku.cn/index.html>) revealed higher levels of SREBP1 pathway-related genes, particularly *SCD1* and *FASN*, in bladder cancer tissues than in normal tissues ($P < 0.05$; Figure 4B). Moreover, patients with lower levels of SREBP1, SCD1, FASN, or ACC presented a significantly greater overall survival rate than those with higher levels of these genes ($P < 0.05$). These findings suggest that targeting SREBP1-mediated lipogenesis is a promising therapeutic strategy for bladder cancer.

Our investigation also revealed that DDH dose-dependently inhibited SREBP1, FASN, ACC, and SCD1 expression while decreasing fatty acid levels in T24, UM-UC-3, and J82 cells (Figure 4C,D). To investigate the role of lipogenesis inhibition in the antiproliferative activity of DDH, we supplemented the experiments with the fatty acids oleic acid (OA) and palmitoleic acid (POA). The results showed that OA and POA significantly attenuated DDH-induced cell death, indicating that DDH suppressed bladder cancer cell proliferation by suppressing SREBP1-mediated fatty acid synthesis (Figure 4E).

GAD improves DDH stability and antiproliferative activity

Anthocyanins and flavonoids are at risk of degradation during processing and storage, and improving their stability is beneficial for the quality and functional properties of DDH. In a previous study, GAD, a phenolic acid (Figure 5A), was identified as a potential stability enhancer with additional pharmacological activities, such as anti-inflammatory, hepatoprotective, and anticancer effects [18,25,26]. We examined whether GAD could improve the stability of anthocyanins and flavonoids in DDH and enhance their anti-bladder cancer effects *in vitro* and *in vivo*. The thermal degradation of TA and TF followed a first-order model in the presence or absence of GAD ($R^2 > 0.9476$; Figure 5B). GAD led to a dose-dependent increase in the $t_{1/2}$ of TA and TF in model solutions, and higher GAD concentrations (100 mg/100 mL) more efficiently stabilized TA and TF, with 16.67% and 15.22% increases in their $t_{1/2}$, respectively (Figure 5B). GAD suppressed bladder cancer cell growth in a dose-dependent manner and potentiated the cytotoxic effects of DDH in T24, UM-UC-3, and J82 cells (Figure 5C,D). Mechanistic studies also revealed that GAD reinforced the inhibitory effect of DDH on the SREBP1 pathway (SREBP1, SCD1, FASN, and ACC expression) in bladder cancer cells (Figure 5E).

In vivo experiments indicated that GAD strengthened the DDH-induced growth suppression of tumor xenografts (Figure 5F). Moreover, we used the coefficient of drug interaction (CDI) to calculate the synergy of the cotreatment [27]. The CDI between DDH and GAD was 0.21, indicating a significant synergistic effect ($CDI < 0.7$). Importantly, the administration of DDH or GAD alone or in combination did not significantly affect the body weight of the mice or the relative organ weights of the liver and kidney, indicating that DDH and GAD were well tolerated in mice alone or in combination (Figure 5F-I). These results suggest that cotreatment with DDH and GAD is a promising strategy for bladder cancer treatment.

Discussion

Bladder cancer is an intractable malignant tumor that severely affects human health and quality of life. Thus, there is a need to develop safe and effective treatments for these patients. Phyto-

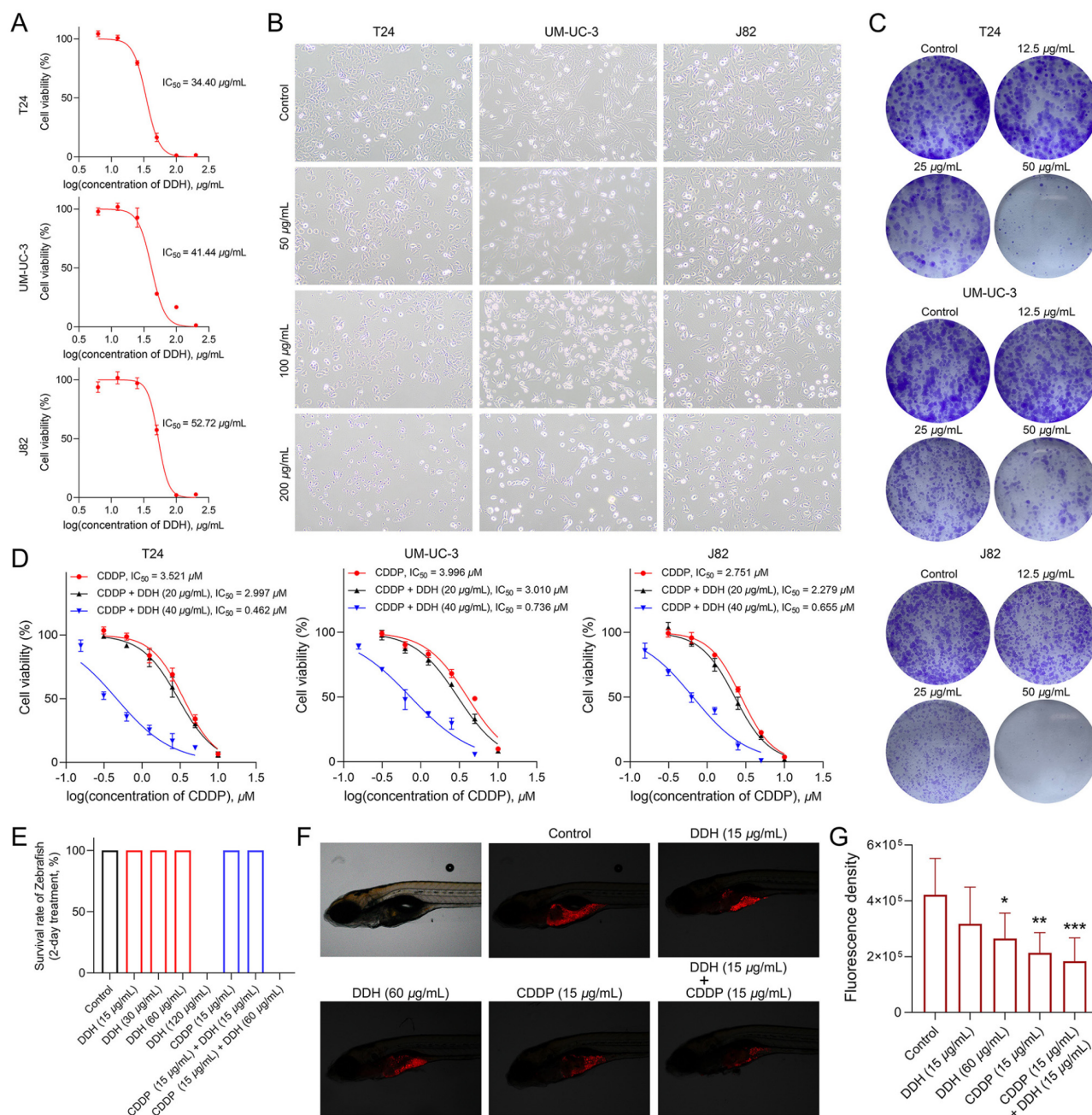
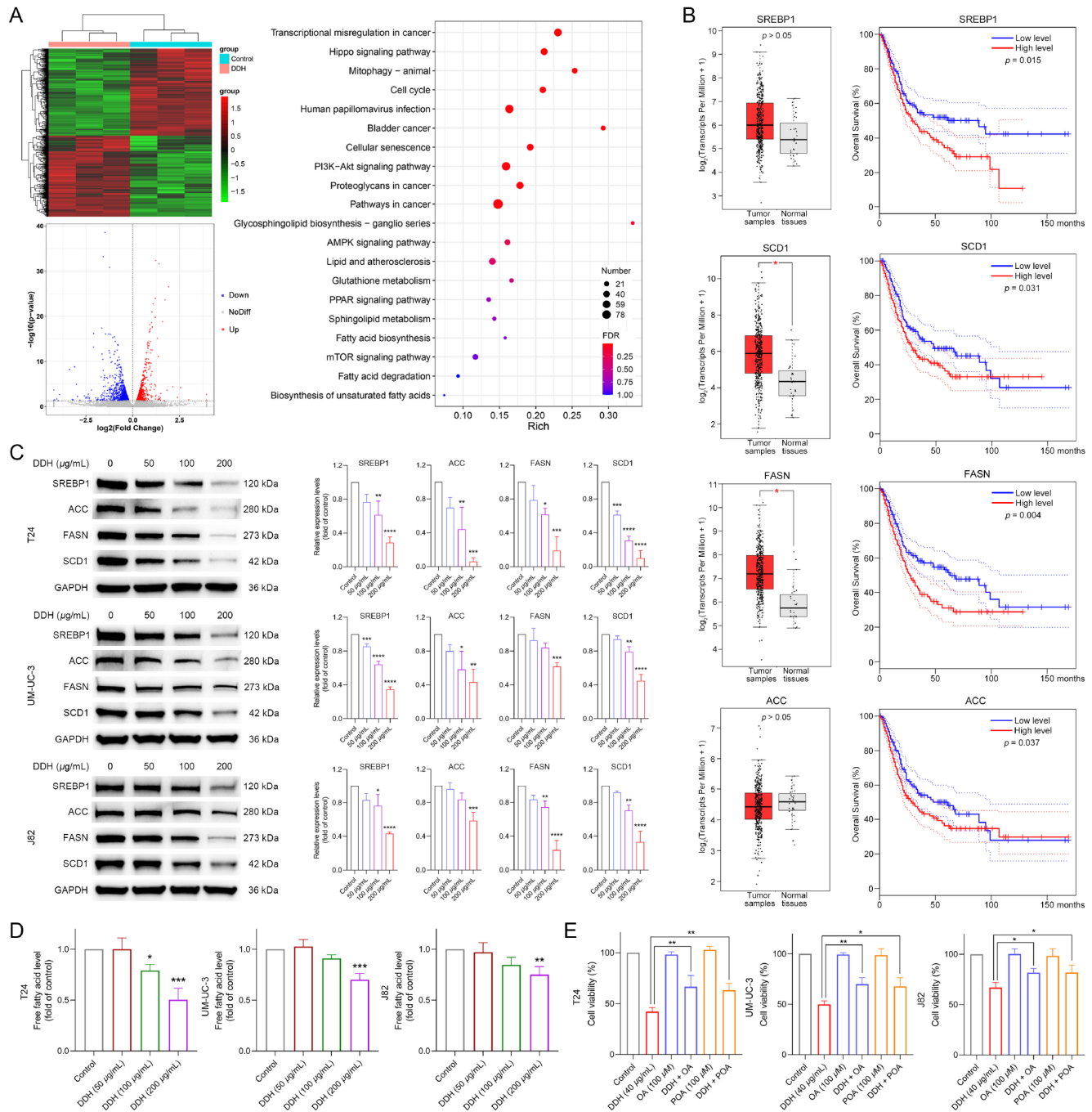


Figure 3. DDH inhibits the proliferation of human bladder cancer cells (A) DDH decreased the viability of T24, UM-UC-3 and J82 cells. (B) DDH affected the morphology of T24, UM-UC-3 and J82 cells. (C) DDH decreased the colony formation of T24, UM-UC-3 and J82 cells. (D) DDH enhanced the cytotoxicity of CDDP against T24, UM-UC-3 and J82 cells. (E) DDH affected the survival rate of zebrafish after 2 days of treatment. (F,G) DDH inhibited bladder cancer cell growth and strengthened CDDP-induced growth suppression in a zebrafish model bearing T24 cells. $n = 3$, * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$ compared with the control group.

chemicals have garnered increasing attention owing to their significant antitumor properties. Our research group has been dedicated to identifying potential anti-bladder cancer agents from medicinal plants in recent years [16,17,28]. In this study, we investigated the anti-bladder cancer effect and underlying mechanism of DDH via *in vitro* and *in vivo* experiments.

Crude extracts of *C. ternatea* flowers have been used to treat arthritis, prevent methylglyoxal-induced glycation and oxidative damage, and reduce oxidative stress and inflammation in obese

individuals [9,10]. The main pharmacological components in *C. ternatea* flowers are flavonoids and anthocyanins [11,13]; however, efficient separation and analytical methods for these compounds are lacking. This study used macroporous resin to separate flavonoids and anthocyanins from *C. ternatea* flowers. We systematically optimized the separation conditions, resulting in a more than 3-fold increase in purity. We also developed an efficient and rapid two-dimensional UPLC-DAD-MS/MS method to determine the flavonoids and anthocyanins present in *C. ternatea*. The two-dimen-



sional UPLC-DAD-MS/MS method revealed 11 anthocyanins and 7 flavonoids within 22 min, and the method was validated to be accurate, stable, and reliable for simultaneous qualitative and quantitative analyses of the flavonoids and anthocyanins present in *C. ternatea* flowers.

The reprogramming of lipid metabolism is among the most prominent metabolic alterations in cancer, and increasing evidence

suggests that lipid metabolism is typically increased at various stages of cancer development [4]. SREBP1 signaling is a key regulator of *de novo* fatty acid synthesis. Our findings indicate a strong association between SREBP1 and its downstream genes (FASN, ACC, and SCD1) and bladder cancer development and prognosis (Figure 4B). Previous studies have reported that inhibiting lipogenesis could effectively suppress bladder cancer cell

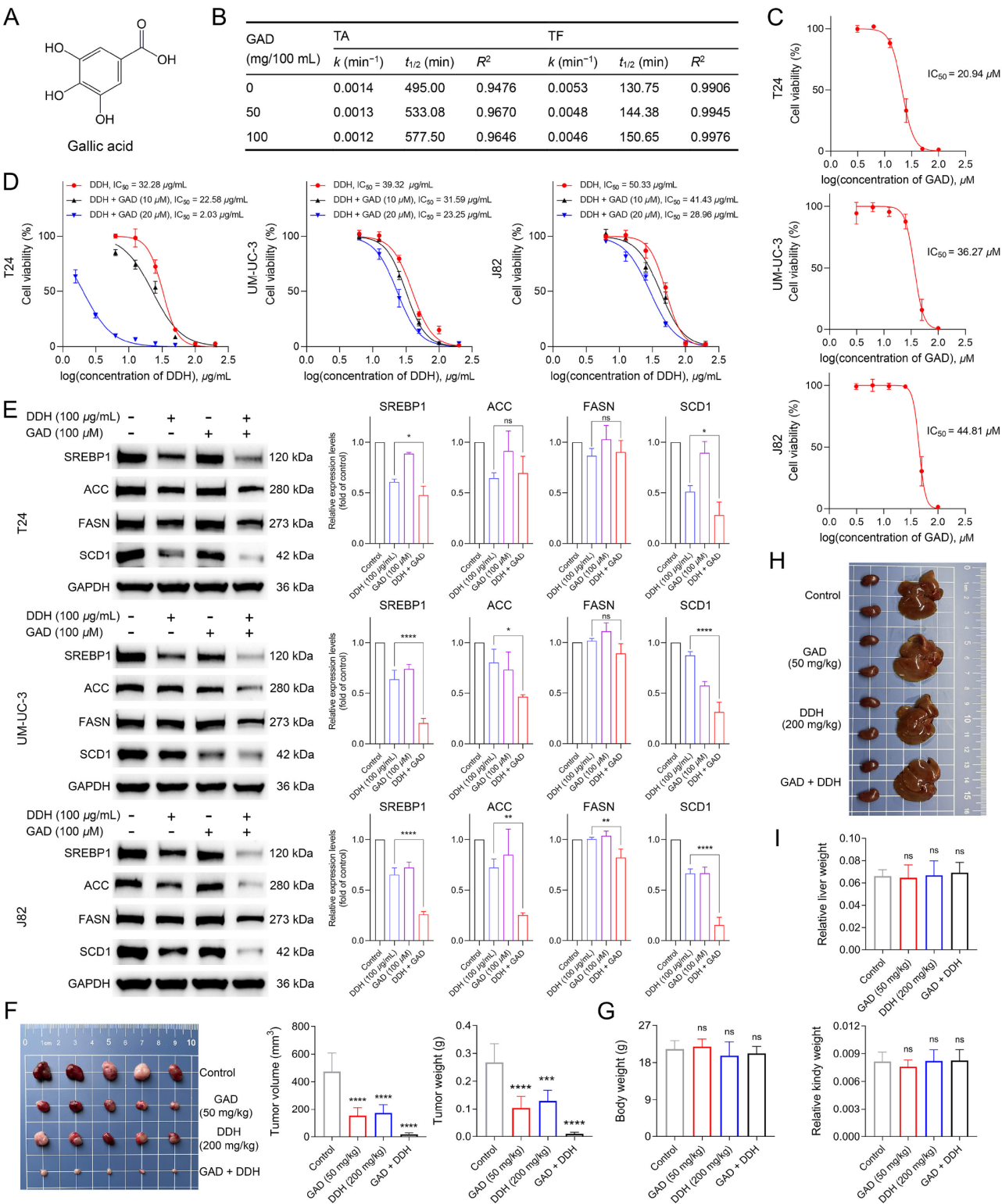


Figure 5. GAD improves DDH stability and antiproliferative activity against bladder cancer (A) The structure of GAD. (B) Degradation kinetics of anthocyanins and flavonoids in DDH with or without GAD. (C) GAD decreased the viability of bladder cancer cells. (D) GAD potentiated the cytotoxic effects of DDH in bladder cancer cells. (E) GAD reinforced the inhibitory effect of DDH on the expressions of SREBP1, ACC, FASN, and SCD1 in bladder cancer cells. (F) DDH and GAD alone or in combination reduced the volume and weight of T24 tumor xenografts, $n = 5$, $***P < 0.001$ and $****P < 0.0001$ compared with the control group. (G) DDH and GAD alone or in combination had no significant effects on the body weight of mice. (H,I) Representative photographs of the liver and kidney in DDH and GAD alone or in combination treated mice and the relative liver and kidney weights in mice, $n = 5$, ns means not significant compared with the control group.

proliferation and overcome chemoresistance by inhibiting SREBP1 and its downstream targets [5–7,29]. Therefore, targeting the SREBP1 pathway to inhibit lipogenesis has emerged as a promising strategy for treating bladder cancer. Our findings revealed that DDH inhibits bladder cancer cell growth *in vitro* and *in vivo*, and a transcriptomics study revealed that DDH can reduce fatty acid synthesis in bladder cancer cells. DDH dose-dependently down-regulated SREBP1, FASN, ACC, and SCD1 expression, and fatty acid supplementation significantly attenuated DDH-induced cell death. These results indicate that DDH suppresses bladder cancer cell growth by inhibiting the SREBP1 pathway. To date, chemotherapeutic agents remain an important choice for cancer treatment, but the development of chemoresistance severely impacts treatment and prognosis. Previous studies have reported that targeting the SREBP1 pathway can effectively overcome CDDP resistance in various cancers, including bladder cancer, non-small-cell lung carcinoma, and oral squamous cell carcinoma [6,30,31]. Our study demonstrated that DDH synergistically potentiates the anti-bladder cancer activity of CDDP *in vitro* and *in vivo*, suggesting that DDH could be an effective chemotherapy sensitizer in cancer. Furthermore, the phenolic acid compound GAD enhances the cytotoxic effects of DDH against bladder cancer cells by reinforcing the inhibitory effect of DDH on the SREBP1 pathway. Therefore, DDH, either alone or in combination with other chemotherapeutic agents, represents a potential treatment for bladder cancer.

In summary, DDH can inhibit the growth of bladder cancer by suppressing the SREBP1 pathway and lipogenesis. In addition, DDH has the potential as a chemosensitizer for bladder cancer therapy. Targeting the SREBP1 pathway could enhance bladder cancer treatment, and *C. ternatea* flower-derived anthocyanins and flavonoids may hold promise as valuable candidates for treating and preventing bladder cancer.

Supplementary Data

Supplementary data is available at *Acta Biochimica et Biophysica Sinica* online.

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Conflict of Interest

The authors declare that they have no conflict of interest.

References

1. Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, Jemal A. Global cancer statistics 2022: globocan estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Canc J Clin* 2024, 74: 229–263
2. Snaebjornsson MT, Janaki-Raman S, Schulze A. Greasing the wheels of the cancer machine: the role of lipid metabolism in cancer. *Cell Metab* 2020, 31: 62–76
3. Röhrig F, Schulze A. The multifaceted roles of fatty acid synthesis in cancer. *Nat Rev Canc* 2016, 16: 732–749
4. Martin-Perez M, Urdiroz-Urricelqui U, Bigas C, Benitah SA. The role of lipids in cancer progression and metastasis. *Cell Metab* 2022, 34: 1675–1699
5. Sugino T, Baba K, Hoshi N, Aikawa K, Yamaguchi O, Suzuki T. Overexpression of fatty acid synthase in human urinary bladder cancer and combined expression of the synthase and Ki-67 as a predictor of prognosis of cancer patients. *Med Mol Morphol* 2011, 44: 146–150
6. Shahid M, Kim M, Jin P, Zhou B, Wang Y, Yang W, You S, *et al.* S-Palmitoylation as a functional regulator of proteins associated with cisplatin resistance in bladder cancer. *Int J Biol Sci* 2020, 16: 2490–2505
7. Du X, Wang Q, Chan E, Merchant M, Liu J, French D, Ashkenazi A, *et al.* FGFR3 stimulates stearoyl CoA desaturase 1 activity to promote bladder tumor growth. *Canc Res* 2012, 72: 5843–5855
8. Adhikari R, Sultana S, Bishayi B. *Clitoria ternatea* flower petals: effect on TNFR1 neutralization via downregulation of synovial matrix metalloproteases. *J Ethnopharmacol* 2018, 210: 209–222
9. Chayaratanasin P, Adisakwattana S, Thilavech T. Protective role of *Clitoria ternatea* L. flower extract on methylglyoxal-induced protein glycation and oxidative damage to DNA. *BMC Complement Med Ther* 2021, 21: 80
10. Wang Y, Liu T, Xie Y, Li N, Liu Y, Wen J, Zhang M, *et al.* *Clitoria ternatea* blue petal extract protects against obesity, oxidative stress, and inflammation induced by a high-fat, high-fructose diet in C57BL/6 mice. *Food Res Int* 2022, 162: 112008
11. Nair V, Bang WY, Schreckinger E, Andarwulan N, Cisneros-Zevallos L. Protective role of ternatin anthocyanins and quercetin glycosides from butterfly pea (*clitoria ternatea* leguminosae) blue flower petals against lipopolysaccharide (LPS)-Induced inflammation in macrophage cells. *J Agric Food Chem* 2015, 63: 6355–6365
12. Escher GB, Wen M, Zhang L, Rosso ND, Granato D. Phenolic composition by UHPLC-Q-TOF-MS/MS and stability of anthocyanins from *Clitoria ternatea* L. (butterfly pea) blue petals. *Food Chem* 2020, 331: 127341
13. Caroline Paz Gonçalves G, Lizandra Gomes Rosas A, Carneiro de Sousa R, Regina Rodrigues Vieira T, César de Albuquerque Sousa T, Ramires T, Ferreira Ferreira da Silveira T, *et al.* A green method for anthocyanin extraction from *Clitoria ternatea* flowers cultivated in southern brazil: characterization, *in vivo* toxicity, and biological activity. *Food Chem* 2024, 435: 137575
14. Lv Y, Liu Z, Jia H, Xiu Y, Liu Z, Deng L. Properties of flavonoids in the treatment of bladder cancer (Review). *Exp Ther Med* 2022, 24: 676
15. Arruda Nascimento E, Lima Coutinho L, Silva CJ, Lima VLAG, Santos Aguiar J. In vitro anticancer properties of anthocyanins: a systematic review. *Biochim Biophys Acta Rev Cancer* 2022, 1877: 188748
16. Sang J, Li W, Diao HJ, Fan RZ, Huang JL, Gan L, Zou MF, *et al.* Jolkinolide B targets thioredoxin and glutathione systems to induce ROS-mediated paraptosis and apoptosis in bladder cancer cells. *Cancer Lett* 2021, 509: 13–25
17. Sang J, Liu CK, Liu J, Luo GC, Zheng WJ, Bai Y, Jiang DY, *et al.* Jolkinolide B synergistically potentiates the antitumor activity of GPX4 inhibitors via

- inhibiting TrxR1 in cisplatin-resistant bladder cancer cells. *Biochem Pharmacol* 2024, 223: 116194
18. Sang J, Zhang Y, Sang J, Li C. Anthocyanins from *Nitraria tangutorum*: qualitative and quantitative analyses, antioxidant and anti-inflammatory activities and their stabilities as affected by some phenolic acids. *Food Measure* 2019, 13: 421–430
19. Sang J, Li B, Huang Y, Ma Q, Liu K, Li C. Deep eutectic solvent-based extraction coupled with green two-dimensional HPLC-DAD-ESI-MS/MS for the determination of anthocyanins from *Lycium ruthenicum* Murr. fruit. *Anal Methods* 2018, 10: 1247–1257
20. Zakaria NNA, Okello EJ, Howes M, Birch-Machin MA, Bowman A. *In vitro* protective effects of an aqueous extract of *Clitoria ternatea* L. flower against hydrogen peroxide-induced cytotoxicity and UV-induced mtDNA damage in human keratinocytes. *Phytother Res* 2018, 32: 1064–1072
21. Terahara N, Saito N, Honda T, Toki K, Osajima Y. Acylated anthocyanins of *clitoria ternatea* flowers and their acyl moieties. *Phytochemistry* 1990, 29: 949–953
22. Kazuma K, Noda N, Suzuki M. Flavonoid composition related to petal color in different lines of *Clitoria ternatea*. *Phytochemistry* 2003, 64: 1133–1139
23. Kazuma K. Malonylated flavonol glycosides from the petals of *Clitoria ternatea*. *Phytochemistry* 2003, 62: 229–237
24. Yi J, Zhu J, Wu J, Thompson CB, Jiang X. Oncogenic activation of PI3K-AKT-mTOR signaling suppresses ferroptosis via SREBP-mediated lipogenesis. *Proc Natl Acad Sci USA* 2020, 117: 31189–31197
25. Chen F, Sang J, Zhang Y, Sang J. Development of a green two-dimensional HPLC-DAD/ESI-MS method for the determination of anthocyanins from *Prunus cerasifera* var. *atropurpurea* leaf and improvement of their stability in energy drinks. *Int J Food Sci Tech* 2018, 53: 1494–1502
26. Ashrafizadeh M, Zarrabi A, Mirzaei S, Hashemi F, Samarghandian S, Zabolian A, Hushmandi K, *et al.* Gallic acid for cancer therapy: molecular mechanisms and boosting efficacy by nanoscopy delivery. *Food Chem Toxicol* 2021, 157: 112576
27. Luchtel RA, Bhagat T, Pradhan K, Jacobs Jr WR, Levine M, Verma A, Shenoy N. High-dose ascorbic acid synergizes with anti-PD1 in a lymphoma mouse model. *Proc Natl Acad Sci USA* 2020, 117: 1666–1677
28. Sang J, Gan L, Zou MF, Lin ZJ, Fan RZ, Huang JL, Li W, *et al.* Jolkinolide B sensitizes bladder cancer to mTOR inhibitors via dual inhibition of Akt signaling and autophagy. *Cancer Lett* 2022, 526: 352–362
29. Deng J, Peng M, Zhou S, Xiao D, Hu X, Xu S, Wu J, *et al.* Metformin targets clusterin to control lipogenesis and inhibit the growth of bladder cancer cells through SREBP-1c/FASN axis. *Sig Transduct Target Ther* 2021, 6: 98
30. Tiong TY, Weng PW, Wang CH, Setiawan SA, Yadav VK, Pikatan NW, Fong IH, *et al.* Targeting the SREBP-1/Hsa-Mir-497/SCAP/FASN oncometabolic axis inhibits the cancer stem-like and chemoresistant phenotype of non-small cell lung carcinoma cells. *Int J Mol Sci* 2022, 23: 7283
31. Almeida LY, Moreira FS, Santos GAS, Cuadra Zelaya FJM, Ortiz CA, Agostini M, Mariano FS, *et al.* FASN inhibition sensitizes metastatic OSCC cells to cisplatin and paclitaxel by downregulating cyclin B1. *Oral Dis* 2023, 29: 649–660